# Business Case

## NovaTech Industries — Intelligent Procurement & Logistics Management Platform

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**Project sponsor:** Director of Operations

**Project owner:** Business Transformation & Analytics Team

> **Project context:** NovaTech Industries is a fictional company created for a hands-on software engineering and AI automation project. Company information, operating metrics, and business assumptions in this document are synthetic. The business case is intentionally written in the style of an internal corporate proposal.

## 1. Executive Summary

NovaTech Industries is a European industrial equipment and electronic components distributor headquartered in the Netherlands, serving manufacturing, engineering, and technology customers across international markets.

As the company expands its supplier network, product portfolio, and distribution operations, the increasing complexity of procurement and logistics activities has highlighted the need for a more integrated, data-driven operating model.

Currently, procurement teams rely on multiple spreadsheets, manual supplier communications, and fragmented operational records to manage purchasing decisions, inventory availability, and shipment tracking. These practices create inefficiencies, limit visibility across departments, and increase the risk of delays, purchasing errors, and unnecessary operational costs.

To address these challenges, NovaTech Industries proposes the development of an **Intelligent Procurement & Logistics Management Platform** to centralize operational information, standardize purchasing processes, automate repetitive activities, and introduce artificial intelligence capabilities to support decision-making.

The initiative will be implemented progressively, beginning with supplier and procurement management before expanding into inventory control, logistics coordination, business intelligence, and AI-powered automation.

## 2. Company Overview

| Attribute | Description |
|---|---|
| Company | NovaTech Industries |
| Headquarters | Rotterdam, Netherlands |
| Industry | Industrial Equipment & Electronic Components |
| Business model | B2B distribution and procurement |
| Geographic operations | Europe, Asia, and North America |
| Supplier network | 50 active suppliers |
| Product portfolio | 200 SKUs |
| Distribution network | 3 warehouses |
| Core departments | Procurement, Logistics, Operations, Finance, Sales |
| Project sponsor | Director of Operations |
| Project owner | Business Transformation & Analytics Team |

NovaTech Industries sources electrical equipment, automation components, mechanical parts, and industrial supplies from international manufacturers and distributors.

Its operations involve supplier negotiations, purchasing, inventory planning, international transportation, warehouse management, and order fulfillment.

## 3. Current Business Challenges

### *3.1 Fragmented supplier information*

Supplier profiles, commercial agreements, product catalogs, and historical performance records are maintained across separate spreadsheets. This limits the procurement team's ability to compare suppliers consistently and identify competitive sourcing options.

### *3.2 Manual purchasing processes*

Purchase requisitions, quotation evaluations, and purchase order approvals require substantial manual intervention, increasing processing times and creating administrative bottlenecks.

### *3.3 Limited inventory visibility*

The absence of a consolidated inventory management system makes it difficult to monitor stock availability, anticipate replenishment requirements, and coordinate purchasing activities across warehouses.

### 3.4 Inconsistent shipment tracking

Logistics teams rely on information received from suppliers and transportation providers to monitor shipments. Delayed updates reduce the company's ability to identify delivery risks and respond proactively.

### 3.5 Limited operational analytics

Procurement, inventory, and logistics data are not consistently integrated into management reporting, making it difficult to evaluate supplier performance, identify cost-saving opportunities, and monitor operational KPIs.

### 3.6 Underutilization of automation and AI

Several recurring administrative activities could be automated. Emerging AI technologies also present opportunities to support supplier evaluation, document processing, exception management, and operational decision-making.

## 4. Project Objectives

The primary objective is to establish a centralized digital platform that improves operational efficiency, increases supply chain visibility, and enables data-driven decision-making.

The following performance improvements are proposed as initial project targets:

| KPI | Baseline | Target |
|---|---|---|
| Average purchase requisition processing time | 3.5 days | 1.5 days |
| On-time supplier delivery rate | 87% | 95% |
| Manual procurement processing time | 100% baseline | 40% reduction |
| Inventory stockout incidents | 24 per quarter | 12 per quarter |
| Average supplier quotation evaluation time | 90 minutes | 20 minutes |
| Purchase order data accuracy | 93% | 99% |

These baseline values are planning assumptions for the simulated company and will be reflected in the synthetic operational datasets. Targets will be used to evaluate the proposed solution.

## 5. Proposed Solution

NovaTech Industries will develop a modular platform comprising six integrated business capabilities.

### 5.1 Supplier Management

A centralized supplier database containing company information, product availability, commercial conditions, lead times, and historical performance.

### 5.2 Procurement Management

A structured process for purchase requisitions, quotation evaluations, supplier selection, approval workflows, and purchase order generation.

### 5.3 Inventory Management

Consolidated visibility into stock levels, inventory movements, reorder points, and replenishment requirements across warehouses.

### 5.4 Logistics Management

Shipment records, transportation milestones, estimated delivery dates, freight costs, and supplier delivery performance.

### 5.5 Business Intelligence

Operational dashboards providing procurement expenditure analysis, supplier scorecards, inventory performance, and logistics KPIs.

### 5.6 Intelligent Automation

Workflow automation and AI-assisted capabilities supporting document processing, supplier comparisons, operational alerts, and purchasing recommendations.

## 6. Data Strategy

The platform will establish a unified operational data model connecting supplier, product, procurement, inventory, and logistics information.

The initial implementation will include the following datasets:

| Business area | Core datasets |
|---|---|
| Supplier Management | Suppliers, Supplier Products |
| Product Management | Products, Product Categories |
| Procurement | Purchase Requisitions, Requisition Items, Quotations, Quotation Items, Purchase Orders, Purchase Order Items |
| Inventory | Warehouses, Inventory Balances, Inventory Movements |
| Logistics | Shipments, Shipment Events |
| Performance Management | Supplier Performance, Operational KPI History |

The data architecture will support referential integrity, historical analysis, business rule validation, and integration with automation and AI components.

For development and testing, synthetic datasets will reproduce realistic business transactions, supplier behaviors, demand fluctuations, inventory movements, and logistics disruptions.

## 7. Implementation Strategy

The project will follow an incremental delivery approach to reduce technical complexity and validate each business capability before expanding the platform.

| Phase | Deliverable | Business outcome |
|---|---|---|
| Phase 1 | Supplier Management & Comparison Engine | Standardized supplier evaluation |
| Phase 2 | Procurement Database & API | Centralized purchasing records |
| Phase 3 | Inventory & Logistics Integration | Improved operational visibility |
| Phase 4 | Web Application & Dashboards | Unified management interface |
| Phase 5 | Workflow Automation | Reduced manual processing |
| Phase 6 | AI Procurement Assistant | AI-supported operational decisions |
| Phase 7 | System Testing & Deployment | Validated integrated platform |

The development team will prioritize a modular architecture, allowing individual capabilities to be developed, tested, and maintained independently.

## 8. Expected Business Benefits

### *8.1 Operational efficiency*

Standardized workflows and automated validations will reduce repetitive administrative tasks and improve purchasing consistency.

### *8.2 Cost optimization*

Structured supplier comparisons and consolidated procurement data will support more competitive sourcing decisions.

### *8.3 Supply chain reliability*

Improved inventory visibility and shipment monitoring will help reduce stockouts and delivery disruptions.

### *8.4 Decision intelligence*

Integrated dashboards and AI-assisted analysis will improve access to operational insights and support more informed management decisions.

## 9. Project Risks and Considerations

The implementation will require careful management of data quality, system integration, business rule consistency, and AI reliability.

Particular attention will be given to ensuring that AI-generated recommendations are explainable, validated against operational data, and subject to appropriate human approval.

The platform will initially operate within a controlled **local development environment**, allowing the project team to evaluate functionality, performance, and system behavior before considering production deployment.

## 10. Success Criteria

The initiative will be considered technically successful when the platform can:

• Maintain consistent supplier, product, procurement, inventory, and logistics records.

• Execute purchasing workflows from requisition through goods receipt.

• Evaluate supplier quotations against defined business constraints.

• Update inventory accurately following approved transactions.

• Generate operational dashboards and performance indicators.

• Execute automated workflows with appropriate exception handling.

• Provide AI-assisted recommendations supported by verifiable business data.

• Operate reliably within the defined development environment.

## 11. Conclusion

The Intelligent Procurement & Logistics Management Platform represents a strategic opportunity for NovaTech Industries to modernize its procurement and supply chain operations.

By consolidating operational information, standardizing processes, and introducing intelligent automation, the company aims to establish a scalable digital foundation that supports operational efficiency, transparency, and continued business growth.

The proposed phased implementation will allow NovaTech Industries to validate business value progressively while establishing the technical capabilities required for more advanced automation and AI-driven decision support.